

Multiscale Spatiotemporal Model of White Matter Demyelination & Remyelination: Equations and Model Parameters

Zheng Xia, Theyn kan, Iga Pregowska

March 16, 2026

Spatial and temporal discretization - *Reaction-Diffusion Systems lecture*

We use a 2D grid of $n_x \times n_y = 50 \times 50$ pixels with pixel size

$$\Delta x = \Delta y = 50 \text{ } \mu\text{m} = 0.05 \text{ mm.}$$

Thus the physical domain size is $2.5 \text{ mm} \times 2.5 \text{ mm}$. The PDE solver uses a timestep $\Delta t_{\text{PDE}} = 0.05 \text{ min}$ and we take 10 PDE substeps per macro step, so

$$\Delta t = \Delta t_{\text{macro}} = 0.5 \text{ min.}$$

Each macro step contains $N_{\text{MCS}} = 40000$ CPM copy attempts. We simulate $n_{\text{macro}} = 500$ macro steps.

Neumann boundary conditions:

$$\nabla I \cdot \mathbf{n}|_{\partial\Omega} = 0 \quad (1)$$

$$\nabla G \cdot \mathbf{n}|_{\partial\Omega} = 0 \quad (2)$$

Reaction-Diffusion - *Reaction-Diffusion Systems lecture*

$$\frac{\partial I}{\partial t} = D_I \nabla^2 I + S_I - k_I I - U_I \quad (3)$$

$$\frac{\partial G}{\partial t} = D_G \nabla^2 G + S_G - U_G - k_G G \quad (4)$$

- **Source Terms:**

$$S_I = s_I \cdot \mathbb{I}_{\text{micro}} \cdot A \quad (5)$$

$$S_G = s_G \cdot \mathbb{I}_{\text{oligo}} \quad (6)$$

- **Uptake Terms:**

$$U_G = \mathbb{I}_{\text{OPC}} \cdot \frac{V_{\text{max}}(\sigma) \cdot G}{K_m \left(1 + \frac{I}{K_i}\right) + G} \quad (7)$$

$$U_I = \eta_I \cdot \mathbb{I}_{\text{OPC}} \cdot I \quad (8)$$

- **Where V_{max} is genotype-dependent:**

$$V_{\text{max}}(\sigma) = \begin{cases} V_{\text{max,mut}} & \text{if genotype} = 1 \\ V_{\text{max,wt}} & \text{otherwise} \end{cases} \quad (9)$$

Table 1: Model Parameters and Configuration Values for Reaction-Diffusion of I and G.

Symbol	Description	Config Value	Units/Notes
<i>Diffusion & Decay</i>			
D_I	Diffusion coefficient of I	0.03	mm^2/min
D_G	Diffusion coefficient of G	0.02	mm^2/min
k_I	Natural decay rate of I	0.0462	$\mu\text{M}/\text{min}$
k_G	Natural decay rate of G	0.0347	$\mu\text{M}/\text{min}$
<i>Source & Production</i>			
s_I	Microglial production rate	0.05 / 0.14	$\mu\text{M}/\text{min}$ Mild/Strong
s_G	Oligodendrocyte production rate	0.06	$\mu\text{M}/\text{min}$
<i>Uptake & Michaelis-Menten</i>			
η_I	OPC clearance rate of I	0.06	—
$V_{\text{max,wt}}$	Max G uptake (Wild Type)	0.12	$\mu\text{M}/\text{min}$
$V_{\text{max,mut}}$	Max G uptake (Mutant)	0.192	$\mu\text{M}/\text{min}$
K_m	Michaelis constant for G	0.50	μM
K_i	Inhibition constant (current)	2.0	Mild: 2.0 / Strong: 0.5

Fourier stability

$$\alpha_x = \frac{D\Delta t}{\Delta x^2}, \quad \alpha_y = \frac{D\Delta t}{\Delta y^2} \quad (10)$$

$$\alpha_x + \alpha_y \leq \frac{1}{2} \quad (11)$$

Cellular Potts Model - *Cell based modeling lecture*

Boundary constraints

$$H_{\text{adhesion}} = \sum_{\langle \vec{i}, \vec{j} \rangle} J(\tau(\sigma_{\vec{i}}), \tau(\sigma_{\vec{j}}))(1 - \delta_{\sigma_{\vec{i}}, \sigma_{\vec{j}}}) \quad (12)$$

Area Constraints

$$H_{\text{area}} = \sum_{\sigma} \lambda_{\text{vol}}(\sigma)(v(\sigma) - V_t(\sigma))^2 \quad (13)$$

Chemotaxis energy

$$H_{\text{chem}} = -\lambda_{\text{chem}} \left[\frac{C_o(\vec{i})}{1 + \frac{C_o(\vec{i})}{C_{\text{sat}}}} - \frac{C_o(\vec{i}')} {1 + \frac{C_o(\vec{i}')}{C_{\text{sat}}}} \right] \quad (14)$$

State Change

$$H_{\text{new}} = H_{\text{area}} + H_{\text{boundary}} + H_{\text{chem}} \quad (15)$$

$$\Delta H = H_{\text{new}} - H_{\text{old}} \quad (16)$$

$$P(\sigma(\vec{i}) \rightarrow \sigma(\vec{i}')) = \begin{cases} e^{-\frac{\Delta H}{T}}, & \text{if } \Delta H > 0 \\ 1, & \text{if } \Delta H \leq 0 \end{cases} \quad (17)$$

Cell Type 1	Cell Type 2	Energy (J)
MEDIUM	OPC	10.0
MEDIUM	OLIGO	12.0
MEDIUM	MICRO	11.0
OPC	OLIGO	6.0
OPC	MICRO	9.0
OLIGO	MICRO	9.0

Table 2: Two-sided Adhesion Energies for Cellular Potts Model.

Table 3: Energy Function Parameters				
Parameter	Description	OPC	OLIGO	MICRO
V_t	Initial Target Area (μm)	45.0	55.0	45.0
λ_{area}	Volume Stiffness	0.8	2.5	2.0
λ_{chem}	Chemotaxis Strength	180.0	—	—
C_{sat}	Saturation Constant	0.5	—	—
T	Simulation Temperature	30.0	30.0	30.0

OPC Growth Dynamics

$$\Delta V_t = \alpha \cdot \left(\frac{G}{K_G + G} \right) \cdot \left(\frac{1}{1 + \frac{I}{K_{I,grow}}} \right) \quad (18)$$

$$V_t(t+1) = \min(V_{div}, V_t(t) + \Delta V_t) \quad (19)$$

Table 4: Cell Growth and Volume Parameters			
Parameter	Symbol	Value	Description
Growth Coefficient	α	3.0	Maximum growth rate
Half-saturation (G)	K_G	0.6	G-concentration for half-max growth
Inhibition Const (I)	$K_{I,grow}$	1.5 / 0.7	Sensitivity to growth inhibition (Mild/Strong)
Baseline Volume	$V_{0,opc}$	45.0	Initial target volume for OPCs μm
Division Factor	-	1.4	Multiplier for division threshold
Division Threshold	V_{div}	63.0	Maximum area before division μm

OPC differentiation - *Gene Regulatory Network modelling lecture*

$$\frac{dq_k}{dt} = \alpha_q \left(\frac{I(\vec{x})^{m_I}}{K_I^{m_I} + I(\vec{x})^{m_I}} \right) - \gamma_q q_k \quad (20)$$

$$\frac{dr_k}{dt} = \alpha_r \left(\beta + \frac{G(\vec{x})^{n_G}}{K_G^{n_G} + G(\vec{x})^{n_G}} \right) \left(\frac{1}{1 + (q_k/K_q)^{h_q}} \right) - \gamma_r r_k \quad (21)$$

$$\frac{dp_k}{dt} = \alpha_p r_k - \gamma_p p_k \quad (22)$$

Differentiation of OPC to oligodendrocytes:

$$\tau_k(t + \Delta t) = 2 \quad \text{if } p_k \geq p_{th} \text{ for duration } \Delta t \cdot \tau_{hold} \quad (23)$$

with mutation probability 0.06.

Table 5: Intracellular GRN Parameters

Parameter	Symbol	Value	Description
<i>Inhibition (q)</i>			
Inhibition Rate (Mild/Strong)	α_q	0.4 / 1.2	Rate of q increase via inflammation $\mu M/min$
Inhibition Decay	γ_q	0.8	Natural decay of state q $\mu M/min$
Inhibition Threshold	K_I, K_q	0.8, 0.8	Half-maximal constants
Inhibition Hill Coeffs	m_I, h_q	2, 2	Cooperativity exponents
<i>Commitment (r)</i>			
Commitment Rate	α_r	1.6	Scaling of differentiation drive $\mu M/min$
Commitment Decay	γ_r	0.7	Decay rate of commitment state $\mu M/min$
Growth Factor Threshold	K_G	0.6	Sensitivity to growth factor G
Growth Factor Hill Coeff	n_G	3	Steepness of G response
Baseline Drive	β	0.07	Signal-independent maturation drive
<i>Phenotype (p)</i>			
Protein Prod. Rate	α_p	1.0	Rate of marker accumulation $\mu M/min$
Protein Decay Rate	γ_p	0.5	Decay rate of marker protein $\mu M/min$
Activation Threshold	p_{th}	0.4	Minimum level for maturation
Hold Duration	τ_{hold}	6 steps	Required time above threshold

Myelin Field Dynamics

Temporal evolution of myelin density M :

$$M_{t+\Delta t} = \text{clip} (M_t + \Delta t [\alpha_{rep} \mathcal{R}(\vec{x}) - \eta_{dem} I(\vec{x}) M_t], 0, 1) \quad (24)$$

Spatial repair influence $\mathcal{R}(\vec{x})$ by local mature oligodendrocytes:

$$\mathcal{R}(\vec{x}) = \max_{\vec{j} \in \mathcal{N}(\vec{x})} \delta_{\tau(\sigma(\vec{j})), 2} \quad (25)$$

Table 6: Myelin Simulation Parameters

Parameter	Symbol	Value
Time Step (Macro)	Δt	0.5 min
Repair Rate	α_{rep}	0.04 min ⁻¹
Demyelination Rate	η_{dem}	0.08 min ⁻¹
Healthy Initial Myelin	M_{init}	1.0
Lesion Initial Myelin	M_{lesion}	0.15